

1 Introduction

In poultry farming, optimizing feeding methods is paramount to ensure the health and productivity of chickens. Understanding the impact of different feeding strategies on chicken weight is essential for farmers to make informed decisions regarding nutrition and management practices. This study employs a factorial design experiment to investigate the effects of various feeding methods on chicken weight. By systematically manipulating factors such as the form of protein, level of protein, and level of fish solubles, we aim to elucidate their influence on chicken growth.

2 Recognition of and Statement of the Problem

The problem at hand revolves around the need to identify the most effective feeding methods to promote optimal chicken growth. Recognizing the significance of poultry farming in meeting global protein demands, it is imperative to address this issue to enhance productivity and profitability in the poultry industry. By optimizing feeding practices, farmers can maximize the efficiency of resource utilization and improve overall farm sustainability. Within this context, we want to answer the following questions:

1. What effect the treatments have on a chicken's weight?
2. Whether or not the treatments act independently of each other?

3 Selection of Response Variable:

The response variable chosen for this study is the weight of chickens, measured in grams. Chicken weight serves as a direct indicator of growth and development, reflecting the effectiveness of feeding methods in meeting nutritional requirements. By focusing on chicken weight as the response variable, we can assess the impact of different feeding strategies on overall growth performance.

4 Choice of Factors, Levels, and Range:

The factors considered in this study include the form of protein (Protein), level of protein (LP), and level of fish solubles (LS). Each factor has multiple levels: Protein (GN=Groundnut=1, SB = Soyabean=-1), LP (L=low=-1, M = medium=0, H = high=1), and LS (F = first level=-1, S = second level=1). The range of each factor encompasses the various combinations of levels, resulting in a comprehensive factorial design that allows for the exploration of main effects and interactions.

Factors	Levels
Form of Protein (Protein)	GN=Groundnut=1, SB=Soyabean=-1
Level of Protein (LP)	L=low=-1, M=medium=0, H=high=1
Level of Fish Solubles (LS)	F=first level=-1, S=second level=1

Table 1: Factors and their Levels

5 Choice of Experimental Design:

A $2 \times 3 \times 2$ factorial design was selected for this study due to its efficiency in examining multiple factors and their interactions simultaneously. By systematically varying the levels of each factor, factorial designs enable a thorough investigation of the effects of feeding methods on chicken weight. This design provides statistical robustness and allows for the identification of significant factors influencing growth performance.

6 Statistical Analysis of the Data

6.1 Plot of data

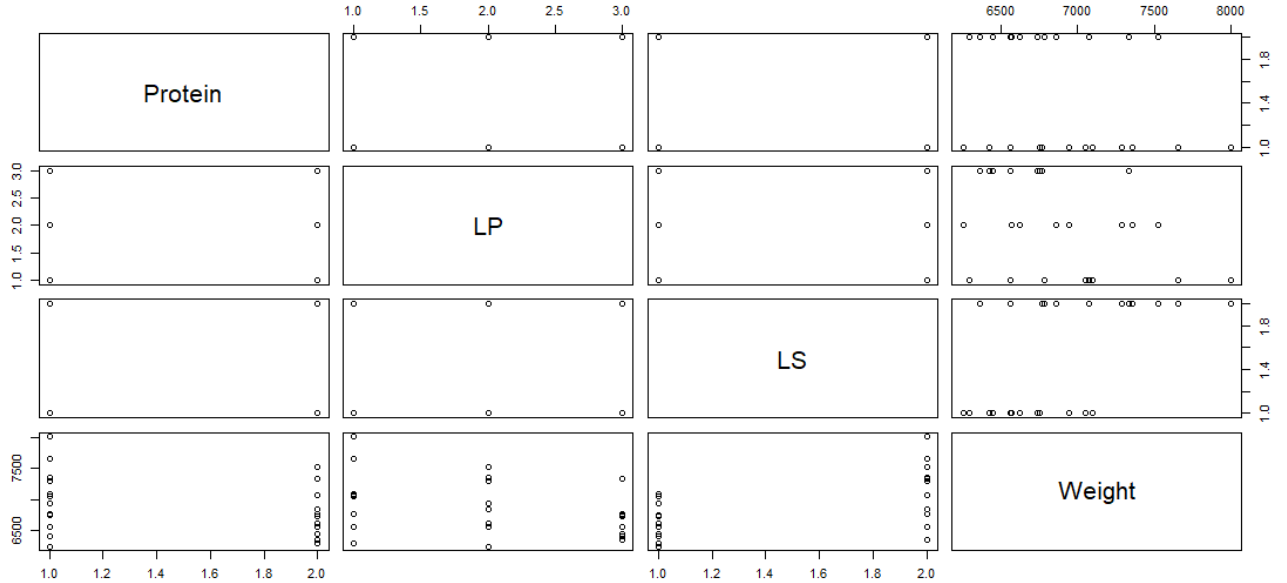


Figure 1: Plot of data

6.2 Interaction Plot

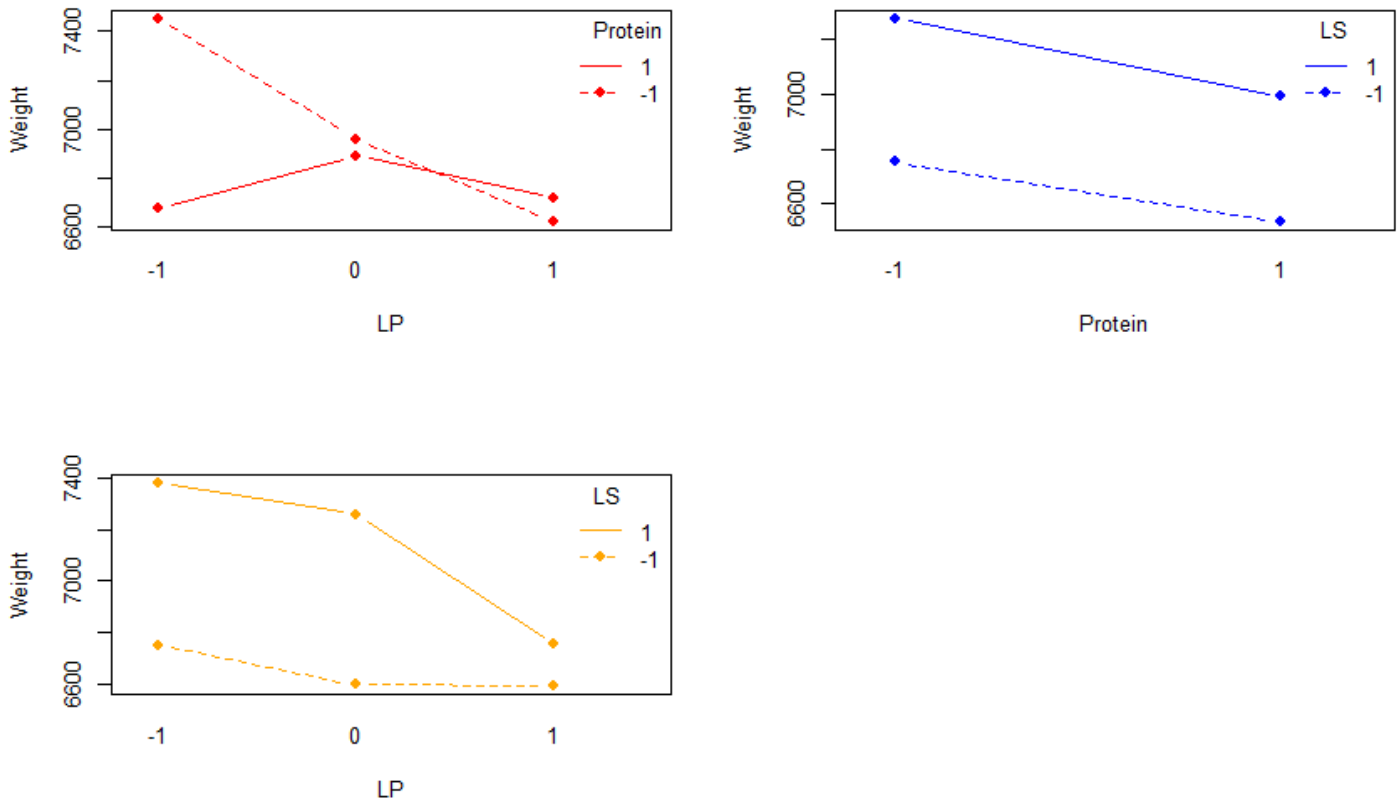


Figure 2: Interaction plot 1

From the plots above, we can see the two lines for protein and level of protein(LP) intersect at a point, so we can say that protein and LP significantly interact to affect the weight of chicken. While the lines for level of fish soluble (LS) are parallel indicating that they do not or will never intersect at any point to affect the weight of chicken, same can be said for level LP and LS. However we need to conduct a statistical test or fit a model to determine which of the factors has effect on chicken weight.

6.3 Boxplot

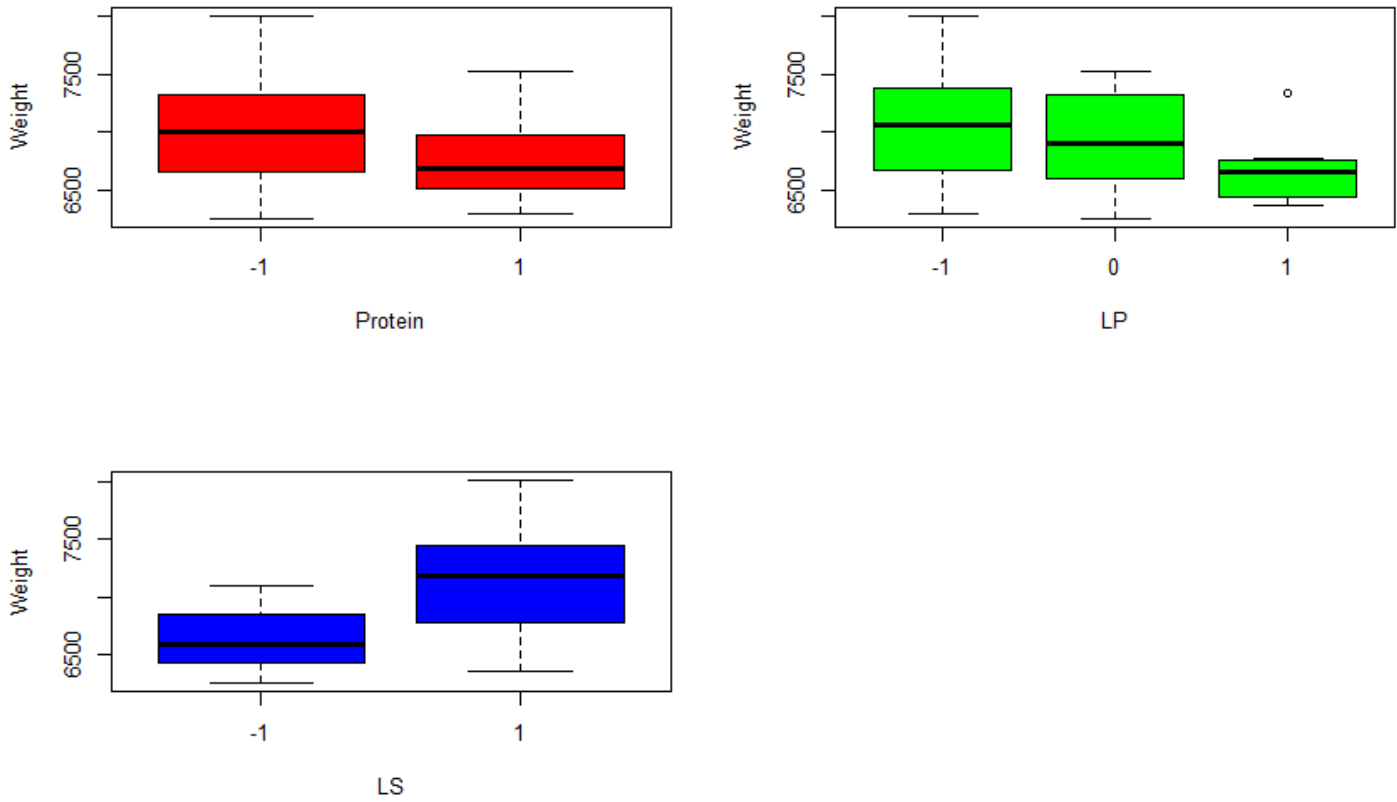


Figure 3: Boxplot plot

From the boxplot above there appear to be no outliers in our data except for one outlying point in the high($H=1$) level of protein (LP).

6.3.1 Checking for Assumptions

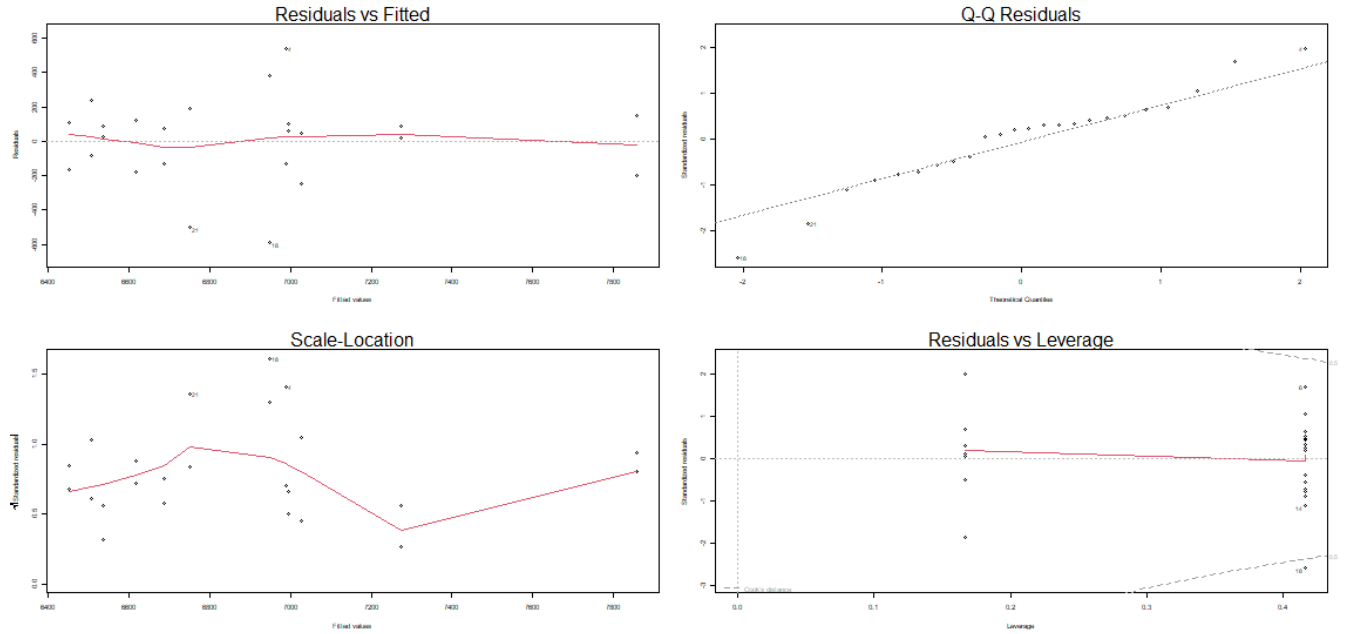


Figure 4: checking assumptions

Analyzing the residuals is an important way to verify the assumptions to determine whether the results from the factorial analysis can be depended upon. From Figure 3 above, the fitted vs. residual plot exhibits a structure-less pattern without any noticeable correlation among the points. The normal probability plot indicates that the residuals closely align with the expected line, with minor deviations with no significant implications. This suggests that concerns regarding normality are unwarranted. In the final plot (Scale-Location), the residuals appear to be distributed uniformly along horizontal lines. Consequently, we can confidently affirm that the assumptions of normality and constant variance are valid.

6.4 Model 1

Call:

```
lm.default(formula = Weight ~ Protein + LP + LS, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-519.79	-191.29	66.25	174.58	552.96

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6887.38	71.35	96.526	< 2e-16 ***
Protein	-124.79	71.35	-1.749	0.09563 .
LP	-196.50	87.39	-2.249	0.03597 *
LS	243.37	71.35	3.411	0.00277 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 349.6 on 20 degrees of freedom

Multiple R-squared: 0.4968, Adjusted R-squared: 0.4214

F-statistic: 6.583 on 3 and 20 DF, p-value: 0.002833

The results of the regression analysis above suggest that weight can be predicted based on Protein, LP, and LS. After adjusting for other variables, the model indicates that while protein show a trend towards decreasing weight, this relationship is not statistically significant. On the other hand, levels of protein are significantly associated with decreased weight, where as levels of fish soluble (LS) are significantly linked with increased weight. The overall model is statistically significant ($F(3, 20) = 6.583$, $p < 0.05$), indicating that the combination of these variables explains a substantial portion, 49.68 percent of the variability in weight. However, the residual standard error of approximately 349.6 suggests that there may be other factors not captured by the model influencing weight outcomes.

6.5 Model 2

Call:

```
lm.default(formula = Weight ~ Protein + LP + LS + Protein:LP +  
            Protein:LS + LP:LS, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-533.29	-135.99	20.23	104.53	539.33

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6887.38	59.88	115.028	< 2e-16 ***
Protein	-124.79	59.88	-2.084	0.052542 .
LP	-196.50	73.33	-2.680	0.015835 *
LS	243.37	59.88	4.065	0.000806 ***
Protein:LP	217.87	73.33	2.971	0.008567 **
Protein:LS	-17.29	59.88	-0.289	0.776232
LP:LS	-115.75	73.33	-1.578	0.132892

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 293.3 on 17 degrees of freedom

Multiple R-squared: 0.6988, Adjusted R-squared: 0.5925

F-statistic: 6.575 on 6 and 17 DF, p-value: 0.00099

From the analysis above, we considered a 2-way interaction effects and can be seen from the output that for the main effect Protein, LP and LS are statistically significant and the interaction between Protein and LP is also significant, the addition of these interaction has increased the adjusted R^2 and R^2 in model 1 from 42.14 to 59.25 and 49.68 to 69.88 percent respectively.

6.6 Model 3

Call:

```
lm.default(formula = Weight ~ Protein + LP + LS + Protein:LP +  
            Protein:LS + LP:LS + Protein:LP:LS, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-587.67	-139.94	51.79	108.63	539.33

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6887.38	60.71	113.443	< 2e-16 ***
Protein	-124.79	60.71	-2.055	0.05653 .
LP	-196.50	74.36	-2.643	0.01773 *
LS	243.37	60.71	4.009	0.00101 **
Protein:LP	217.87	74.36	2.930	0.00981 **
Protein:LS	-17.29	60.71	-0.285	0.77944
LP:LS	-115.75	74.36	-1.557	0.13910
Protein:LP:LS	54.38	74.36	0.731	0.47519

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 297.4 on 16 degrees of freedom

Multiple R-squared: 0.7086, Adjusted R-squared: 0.5811

F-statistic: 5.558 on 7 and 16 DF, p-value: 0.00219

The extended model, which includes the three main factors Protein, LP, LS alongside its three-way interaction term indicates that while Protein, LP, and LS significantly influence weight and also significant interactions exist between Protein and LP. However, the interaction between Protein and LS, LP and LS, and Protein, LP, and LS together are not significant. The model, with an R value of 0.7086, explains approximately 70.86% of weight variability. Overall, the model is statistically significant (F-statistic: 5.558, p-value: 0.00219), suggesting its efficacy in explaining weight variation.

6.7 Comparison of Model 1, 2 and 3

Model 1: $\text{Weight} \sim \text{Protein} + \text{LP} + \text{LS}$

Model 2: $\text{Weight} \sim \text{Protein} + \text{LP} + \text{LS} + \text{Protein:LP} + \text{Protein:LS} + \text{LP:LS}$

Model 3: $\text{Weight} \sim \text{Protein} + \text{LP} + \text{LS} + \text{Protein:LP} + \text{Protein:LS} + \text{LP:LS} + \text{Protein:LP:LS}$

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	20	2443773				
2	17	1462716	3	981057	3.6967	0.03401 *
3	16	1415410	1	47306	0.5348	0.47519

From the results, Model 1 has the highest residual sum of squares (RSS), indicating that it explains the least amount of variation in the data. When we add the interaction terms in Model 2, the RSS decreases significantly ($p=0.03401$), indicating that the interaction terms improve the model fit. However, when we add the three-way interaction term in Model 3, the decrease in RSS is not statistically significant, indicating that the three-way interaction does not significantly improve the model fit compare to model 2. Therefore, Model 2 may be preferred as it achieves a similar explanatory power with fewer parameters. Hence looking at Model 2, the main effect and the interaction term(Protein:LP) appear to statistically significant, so these factors will be Incorporated in our final and ideal model.

6.8 Final Model

Call:

```
lm.default(formula = Weight ~ Protein + LP + LS + Protein:LP,  
            data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-666.33	-135.11	40.62	121.15	522.04

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6887.38	60.77	113.326	< 2e-16 ***
Protein	-124.79	60.77	-2.053	0.054066 .
LP	-196.50	74.43	-2.640	0.016145 *
LS	243.37	60.77	4.005	0.000758 ***
Protein:LP	217.87	74.43	2.927	0.008647 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 297.7 on 19 degrees of freedom

Multiple R-squared: 0.6532, Adjusted R-squared: 0.5802

F-statistic: 8.947 on 4 and 19 DF, p-value: 0.0003077

$$\hat{Y} = 6887.38 - 124.79Protein - 196.50LP + 243.37LS + 217.87Protein : LP$$

This will be our ideal model or best model for this particular analysis, because we have all the factors to be statistically significant and the interaction of protein and LP to be significant. The R-squared value provides insight into how much of the total variability in the response variable is accounted for by the regression model, In this analysis, approximately 65.32 percent of the variation is explained by the model's chosen factors. This substantial percentage suggests that the model effectively captures most of the underlying sources of variation within these factors, instilling confidence in its predictive capability.

Additionally the overall p-value of the anova (F=8.947,p=0.0003077) is less than alpha value=0.05, so we reject the null hypothesis that. the mean levels are same for each treatment or factor. However, this does not tell us which treatment levels are different from each other. It simply tells us that not all of the treatment means are equal.

Hence, we will perform pairwise t-tests using tukey test or Bonferroni's correction for the p-values to calculate pairwise differences between the levels of each treatment.

6.8.1 Anova for Selected Factorial model

Analysis of Variance Table

Response: Weight

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Protein	1	373751	373751	4.2163	0.0540656	.
LP	1	617796	617796	6.9693	0.0161454	*
LS	1	1421553	1421553	16.0364	0.0007583	***
Protein:LP	1	759512	759512	8.5680	0.0086473	**
Residuals	19	1684261	88645			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The ANOVA table reveals significant factors and their p-values. LP and LS are found to be statistically significant predictors of weight, suggesting they have significant effects. Additionally, the interaction term between Protein and LP (Protein:LP) is also significant, indicating a combined influence. However, Protein alone shows marginal significance. Overall, the ANOVA table confirms that both main effects (Protein, LP, LS) and the interaction term (Protein:LP) are significant factors affecting weight, with LS having the most significant impact.

we can proceed to perform post-hoc comparison to determine which treatment levels have different means.

6.8.2 Homogeneity of variance

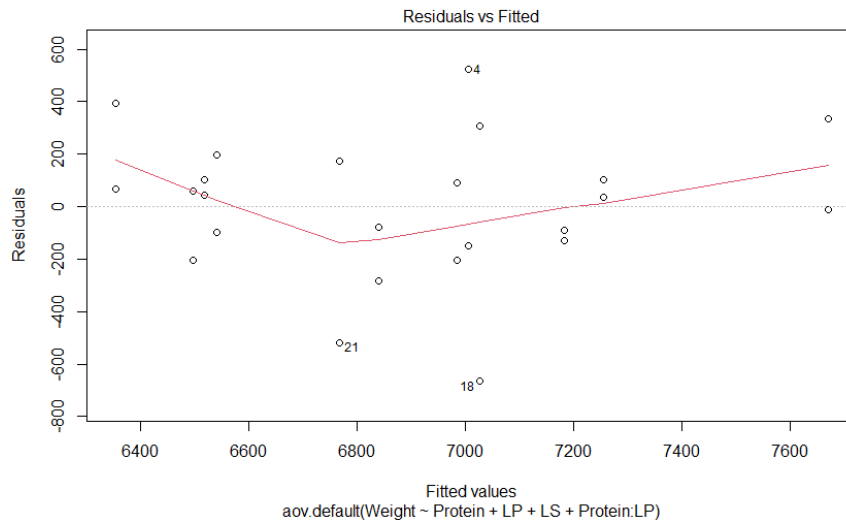


Figure 5: Homogeneity of variance

The residual points are randomly scattered and fall within a horizontal band centered around the 0 line, displaying no systematic pattern of positive and negative values. This shows that the assumption of constant variances is valid.

6.8.3 Normality Plot

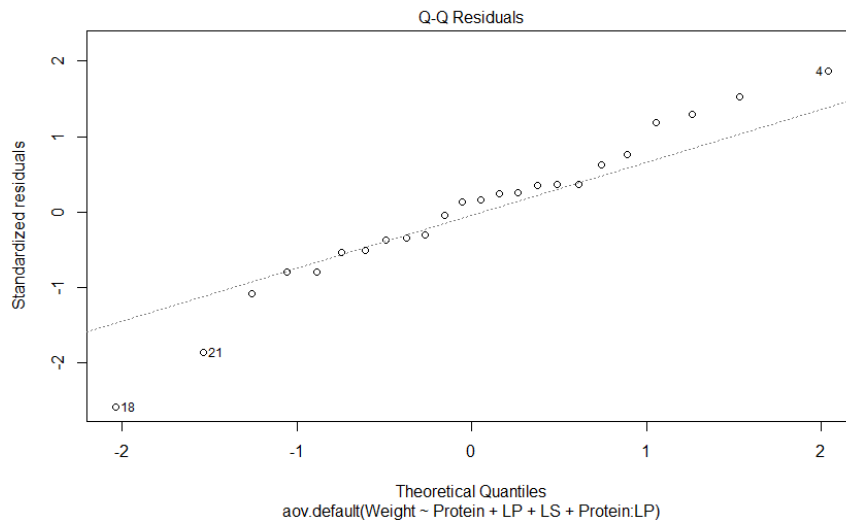


Figure 6: Normality plot

In terms of normality, the points on the normal probability Q-Q plot fall approximately on a straight line aside from the outlying points. This is an indication that the error terms are approximately normally distributed.

6.8.4 Tukey Test

Tukey multiple comparisons of means
99% family-wise confidence level

Fit: aov.default(formula = Weight ~ Protein + LP + LS + Protein * LP, data = data)

\$Protein

	diff	lwr	upr	p adj
1--1	-249.5833	-608.8238	109.6571	0.0601659

\$LP

	diff	lwr	upr	p adj
0--1	-137.625	-646.6308	371.3808	0.6436062
1--1	-393.000	-902.0058	116.0058	0.0477582
1-0	-255.375	-764.3808	253.6308	0.2404547

\$LS

	diff	lwr	upr	p adj
1--1	486.75	127.5096	845.9904	0.0010863

\$'Protein:LP'

	diff	lwr	upr	p adj
1:-1--1:-1	-776.00	-1635.0292	83.02923	0.0219535
-1:0--1:-1	-491.50	-1350.5292	367.52923	0.2508919
1:0--1:-1	-559.75	-1418.7792	299.27923	0.1488021
-1:1--1:-1	-828.75	-1687.7792	30.27923	0.0133403
1:1--1:-1	-733.25	-1592.2792	125.77923	0.0326891
-1:0-1:-1	284.50	-574.5292	1143.52923	0.7678036
1:0-1:-1	216.25	-642.7792	1075.27923	0.9091954
-1:1-1:-1	-52.75	-911.7792	806.27923	0.9998526
1:1-1:-1	42.75	-816.2792	901.77923	0.9999477
1:0--1:0	-68.25	-927.2792	790.77923	0.9994808
-1:1--1:0	-337.25	-1196.2792	521.77923	0.6267715
1:1--1:0	-241.75	-1100.7792	617.27923	0.8639420
-1:1-1:0	-269.00	-1128.0292	590.02923	0.8052182
1:1-1:0	-173.50	-1032.5292	685.52923	0.9620703
1:1--1:1	95.50	-763.5292	954.52923	0.9973976

From the Tukey result:

looking at the Protein results, the difference between the means of the two protein levels is -249.5833. However, the adjusted p-value is 0.0601659, which is greater than 0.05, so we fail to reject the null hypothesis that the means are the same.

In the LP results, for the comparison between 0-1, 1-1 and 0-1, all have an adjusted p-value which is greater than 0.01. So, we fail to reject the null hypothesis

that the means are the same.

In the LS results, the difference between the means in 1 and -1 is 486.75 and the adjusted p-value is 0.0010863, which is less than 0.05. So, we reject the null hypothesis and conclude that there is a significant difference between the means.

In the Protein:LP results, there are several comparisons. For example, for the comparison between 1:-1-1:-1 and -1:0-1:-1, the adjusted p-value is 0.0219535, which is greater than 0.01, so we fail to reject the null hypothesis that the means are the same.

Based on the output, we can see that the only significant difference is in LS between 1 and -1

6.8.5 Tukey Test Plot

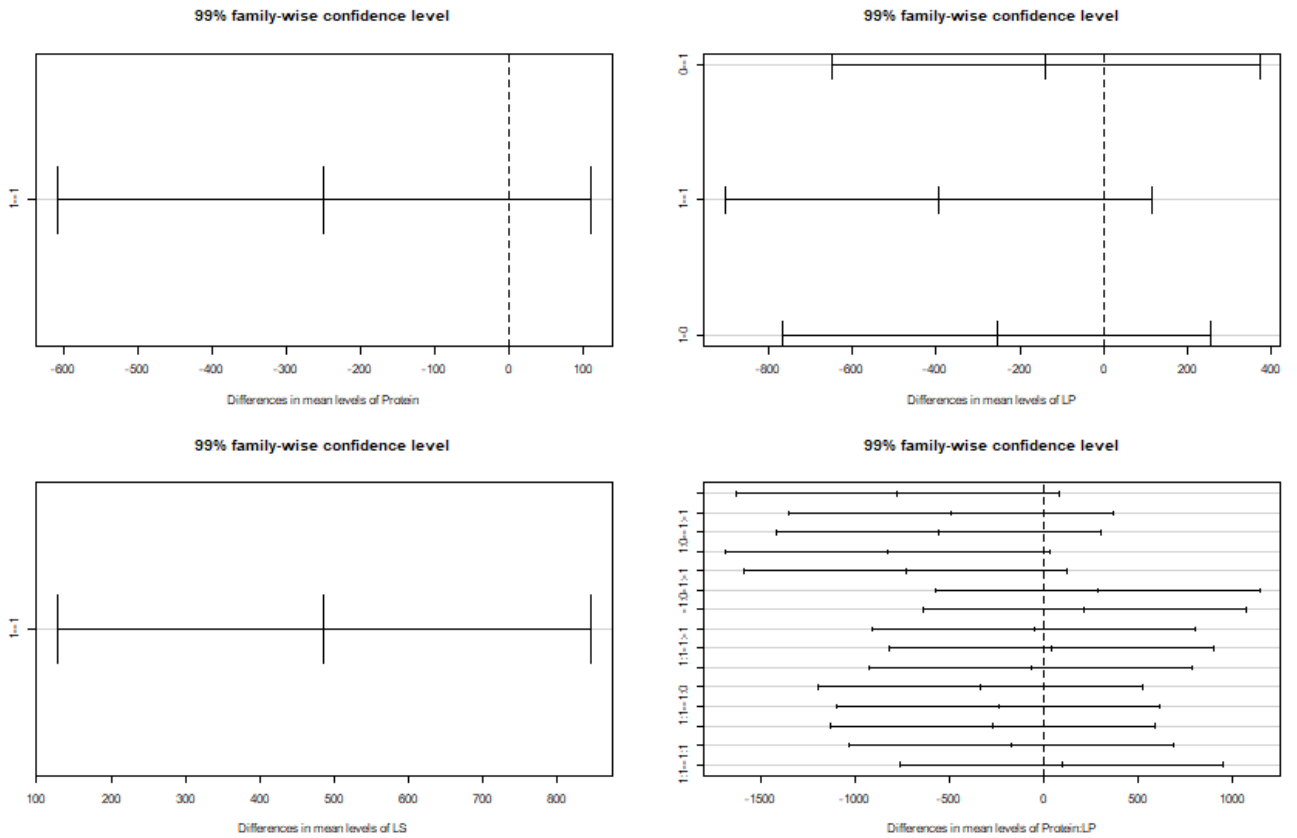


Figure 7: Tukey plot

6.8.6 Bonferroni's Test

Pairwise comparisons using t tests with pooled SD

data: data\Weight and data\Protein

-1

1 0.19

P value adjustment method: bonferroni

Pairwise comparisons using t tests with pooled SD

data: data\Weight and data\LS

-1

1 0.0063

P value adjustment method: bonferroni

Pairwise comparisons using t tests with pooled SD

data: data\$Weight and data\$LP

-1 0

0 1.00 -

1 0.28 0.80

P value adjustment method: bonferroni

Pairwise comparisons using t tests with pooled SD

data: data\$Weight and data\$Protein:data\$LP

-1:-1 -1:0 -1:1 1:-1 1:0

-1:0 1.00 - - -

-1:1 0.15 1.00 - - -

1:-1 0.22 1.00 1.00 - -

1:0 1.00 1.00 1.00 1.00 -

1:1 0.30 1.00 1.00 1.00 1.00

P value adjustment method: bonferroni

The adjusted p-value for the mean difference in Protein between 1 and -1 is 0.19.

The adjusted p-value for the mean difference in level of protein between 1 and -1 is 0.28.

The adjusted p-value for the mean difference in level of protein between 0 and 1 is .08

The adjusted p-value for the mean difference in level of fish soluble between -1 and -1 is 0.0063.

The adjusted p-value for the mean difference in the interaction term Protein: between 1:1 and -1:1 is 0.15

Overall based on the adjusted p-value or output the Bonferroni's test also confirms that the only significant difference is in the LS between 1 and -1.

7 Conclusion

In this study, our focus was to determine or find out if Protein, LP and LS have any effect on chicken weight and asses whether or not these factors act independently of each other whiles ensuring robust statistical outcomes. To achieve this, a $2 \times 3 \times 2$ Factorial Design (FD) was employed as our experimental framework.

Analyzing the outputs in R provided several comprehensive analyses such as,ANOVA, interaction plot, normality plot of residuals,residuals vs predicted plot, comparison of different factorial models, significant relationships between treatments and chicken weight were identified. Specifically, the factorial analysis highlights the significant main effects of Protein LP, and LS, as well as the interaction term between Protein and LP, indicating their influence on chicken weight. Regarding the independence of treatments,given that the interactions between LS and LP, as well as between Protein and LS, are not significant, it implies that these treatments may act independently of each other. Therefore, while Protein and LP may show some dependency due to their significant interaction, LS appears to act independently in influencing chicken weight. Additionally ,the highest weight mean is 7831, which is obtained by combining soybean,low level of protein and Second level of the fish solubles.

8 Recommendations

Further Exploration of Treatment Interactions: While this study has identified significant effects of Protein, LP and LS on chicken weight, the lack of significance in interactions between LS and LP, as well as between Protein and LS, prompts further investigation. Future research could delve deeper into these interactions, employing larger sample sizes.

Consideration of Additional Factors: While this study focused on protein, LP, and LS, other factors such as diet composition, housing conditions, and genetic factors may also influence chicken weight. Integrating these variables into future research designs would offer a more comprehensive understanding of the multifaceted determinants of poultry growth

References

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